

LEAST SQUARES - Intuition

$$\begin{matrix} m \times 1 & & m \times n & & n \times 1 \\ & \nearrow & \uparrow & \nwarrow & \\ \bar{y} & = & A \bar{x} & & \end{matrix}$$

$m > n$
Tall matrix

$$\arg \min. = \| \bar{y} - A \bar{x} \|^2$$

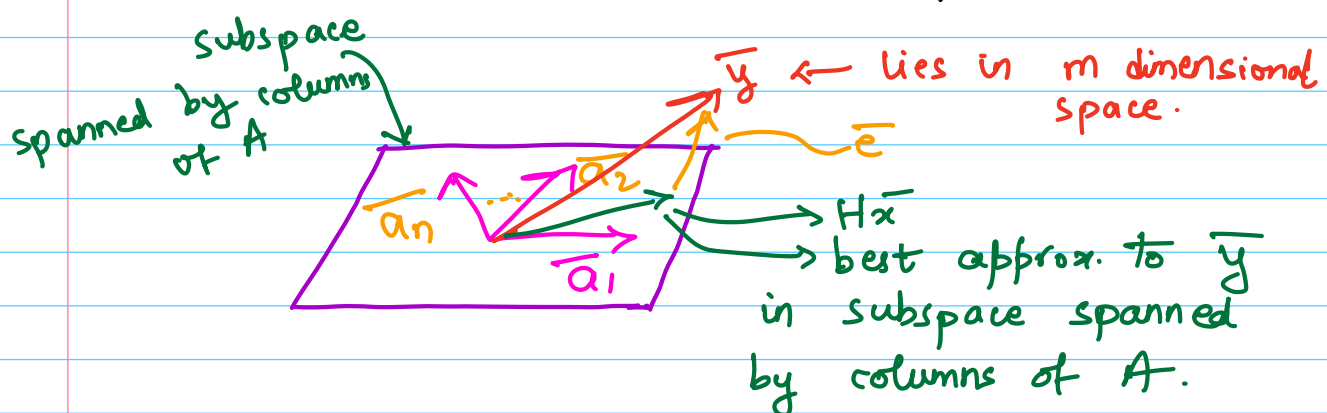
$$\bar{x} = \underbrace{(A^T A)^{-1} A^T}_{\text{pseudo-inv. of } A} \bar{y}$$

$$\bar{y} = \begin{bmatrix} \bar{a}_1 & \bar{a}_2 & \dots & \bar{a}_n \end{bmatrix} \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_n \end{bmatrix}$$

columns of A
 n columns of size m

$$A \bar{x} = \bar{a}_1 \bar{x}_1 + \bar{a}_2 \bar{x}_2 + \dots + \bar{a}_n \bar{x}_n$$

Lies in the column space of A .



$$\bar{e} = \bar{y} - H \bar{x}$$

When is $\bar{e}$ minimum?

$\bar{e}$ is minimum, when it is $\perp^r$ to the span of columns of A

$\Rightarrow \bar{e}$ is orthogonal to every column of matrix A .

$\Rightarrow$ $\left\{ \begin{array}{l} \bar{a}_1^T \bar{e} = 0 \\ \bar{a}_2^T \bar{e} = 0 \\ \vdots \\ \bar{a}_n^T \bar{e} = 0. \end{array} \right.$

n orthogonality conditions.

$\Rightarrow \left[\begin{array}{c} \bar{a}_1^T \\ \bar{a}_2^T \\ \vdots \\ \bar{a}_n^T \end{array} \right] \bar{e} = 0$

A^T

Principle of orthogonality $\Rightarrow A^T \bar{e} = 0$

True for any matrix $A \Rightarrow A^T (\bar{y} - A\bar{x}) = 0$

$\Rightarrow A^T \bar{y} = A^T A \bar{x}$

Normal eqns. $\Rightarrow \bar{x} = (A^T A)^{-1} A^T \bar{y}$

Least Squares Soln.

only if $(A^T A)^{-1}$ is invertible

$\Rightarrow$ Holds if A is full column rank.

For complex matrix A

$$\bar{x} = (A^H A)^{-1} A^H \bar{y}$$

What is $A\bar{x}$?

$$A\bar{x} = A(A^T A)^{-1} A^T \bar{y}$$

Best approx. of $\bar{y}$ in subspace spanned by columns of A

projection of $\bar{y}$ in subspace spanned by columns of A .

$$\Rightarrow A(A^T A)^{-1} A^T \bar{y}$$

yields the projection of $\bar{y}$ in the subspace spanned by columns of A .

$$\Rightarrow A(A^T A)^{-1} A^T = P_A$$

= PROJECTION MATRIX for subspace spanned by columns of A .

$$P_A \cdot \bar{y}$$

projection of $\bar{y}$ in span of $\bar{a}_1, \bar{a}_2, \dots, \bar{a}_n$

$$\begin{aligned} P_A \cdot P_A &= A(A^T A)^{-1} A^T \cdot A(A^T A)^{-1} A^T \\ &= A(A^T A)^{-1} A^T \\ &= P_A. \end{aligned}$$

$$\therefore P_A \cdot P_A = P_A$$

$$\Rightarrow P_A^n = P_A \quad n=1, 2, \dots$$

IDEMPOTENT matrix

$P_A \cdot \bar{y} = \text{projection of } \bar{y}$

$P_A \cdot (P_A \cdot \bar{y}) = \text{projection of}$
 $\text{projection of } \bar{y}$

$$\Rightarrow P_A \cdot P_A \bar{y} = P_A \bar{y} \quad \forall \bar{y}$$

$$\Rightarrow P_A^2 = P_A.$$